

Mini Project 02 – BlazeDemo

Introduction

In this BlazeDemo which is used to purchase flight tickets from two cities. It contains Home page, flight choosing page, reservation page and receipt page. We are going to perform the load test, stress test, spike test, endurance test, and volume test to be here.

- Web application at **<https://www.blazedemo.com>**
- Page load times should be under 2 seconds for 90% of requests.
- Response time should be less than 3 seconds for 95% of the requests.

Test Setup

BlazeDemo

- | | | |
|--|---|-------------------|
| 1. https://www.blazedemo.com/ | - | Home Page |
| 2. https://www.blazedemo.com/reserve.php | - | Reservation Page |
| 3. https://www.blazedemo.com/purchase.php | - | Purchase Page |
| 4. https://www.blazedemo.com/confirmation.php | - | Confirmation Page |

Test-Setup

1. From blaze meter to get the data from website.
2. Using Synchronizing Timer to integrate no of user requests.
3. Using Random Controller to process the requests on randomly.
4. Using Loop Controller in infinite mode to repeat the activity.
5. The above features is integrate with multiple thread group by duration to stop specify thread lifetime according to the thread group.

Environment Setup

1. **Processor** - 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz
2. **RAM** – 8GB

Jenkins

It integrated with Jenkin's by running with the local environment.

GitHub

1. https://github.com/senthilraja29/Performance_Testing/blob/master/Blazedemo%20Endurance%20test.jmx
2. https://github.com/senthilraja29/Performance_Testing/blob/master/Blazedemo%20Scalability%20test.jmx
3. https://github.com/senthilraja29/Performance_Testing/blob/master/Blazedemo%20Spike%20test.jmx
4. https://github.com/senthilraja29/Performance_Testing/blob/master/Blazedemo%20Stress%20test.jmx
5. https://github.com/senthilraja29/Performance_Testing/blob/master/Blazedemo%20load%20test.jmx

Test Results

Load Test

In this load test to perform 200 concurrent users hitting at 30 mints. After conducting load test around 1.3 lakh users hit the website within 30 mints, we received **7.67%** of error reflects it's around 10k errors. Average response time performed by 983ms and maximum request processed by 411672ms in purchase page. Average throughput time is 66.24 seconds.

Requests		Executions			Response Times (ms)						Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	134459	10319	7.67%	983.11	12	411672	877.00	3757.90	4932.95	6902.96	66.24	385.29	54.50
Confirmation 03	11492	902	7.85%	996.08	21	21990	887.00	1321.70	1581.00	4594.33	5.66	31.02	5.30
Purchase 01	11364	873	7.68%	995.06	16	10613	894.00	1336.00	1574.00	4487.90	5.62	35.67	4.69
Purchase 02	11294	874	7.74%	1025.48	12	411672	891.00	1330.50	1560.25	4599.50	5.61	35.56	4.64
Home page 01	11234	887	7.90%	964.97	18	24396	864.00	1294.00	1523.00	4474.05	5.56	25.16	4.07
Home page 03	11203	832	7.43%	968.09	14	21762	872.00	1304.00	1537.00	4487.00	5.53	25.07	4.06
Purchase 03	11188	863	7.71%	987.94	13	24623	891.50	1314.10	1548.00	4361.39	5.51	34.96	4.58
Confirmation 02	11179	852	7.62%	970.24	19	8126	879.00	1303.00	1529.00	4477.60	5.53	30.25	5.16
Home page 02	11171	840	7.52%	964.63	17	21482	864.00	1295.00	1536.40	4441.96	5.52	25.01	4.05
Reservation 03	11094	866	7.81%	978.87	13	8895	883.00	1311.00	1549.25	4508.05	5.48	38.01	4.33
Confirmation 01	11091	845	7.62%	986.81	22	10580	885.00	1312.00	1563.60	4636.52	5.49	30.09	5.17
Reservation 02	11085	855	7.71%	985.17	14	21598	883.00	1322.00	1549.70	4384.24	5.48	37.91	4.32
Reservation 01	11064	830	7.50%	972.96	15	21592	879.00	1296.50	1529.00	4364.90	5.49	37.91	4.32

Row Labels	Average of Latency	StdDev of Latency	in of Laten	Max of Latency
Confirmation 01	880.805228	589.0571153	0	10580
Confirmation 02	871.8380513	562.859147	0	8126
Confirmation 03	888.8674368	657.9785775	0	21990
Home page 01	861.4611639	641.285284	0	24387
Home page 02	861.1723676	589.4187831	0	21482
Home page 03	865.8711005	612.5830331	0	22174
Purchase 01	880.6361963	595.4096037	0	13022
Purchase 02	899.1879751	3170.708612	0	411648
Purchase 03	878.0877549	615.3723165	0	24602
Reservation 01	860.1158312	623.4326412	0	21581
Reservation 02	866.6775655	625.0247861	0	21598
Reservation 03	863.2038376	574.3029531	0	8895
Grand Total	873.2024018	1086.761787	0	411648

Observations:

1. 10 thousand errors encountered during load testing which is **7.67%**.
2. Average Response time **983.11ms**
3. 95th percentile Response times - **4932.95ms**
4. Throughput (requests per second) - **66.24ms**
5. Network performance (Max latency) – **411648ms**
6. Network performance (Average latency) – **411648ms**

Stress Test

In this stress test, which is conducted by stepping thread groups to perform 500 users holds for 180 seconds. In this load, 50 threads are gradually added by each 20-second interval.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	3055	305	9.98%	893.26	0	2502	962.00	1336.40	1486.40	2332.00	7.44	41.93	5.80
Confirmation 01	243	27	11.11%	899.66	0	2489	975.00	1346.80	1459.40	2328.48	0.59	3.13	0.52
Confirmation 02	245	21	8.57%	962.14	0	2495	1009.00	1440.60	1513.90	2485.86	0.60	3.20	0.54
Confirmation 03	263	19	7.22%	912.57	0	2501	947.00	1315.80	1492.00	2486.56	0.64	3.46	0.59
Home page 01	279	28	10.04%	868.51	0	2357	918.00	1314.00	1463.00	2302.00	0.68	3.00	0.47
Home page 02	255	28	10.98%	892.23	0	2478	976.00	1335.00	1467.20	2015.24	0.62	2.73	0.43
Home page 03	251	31	12.35%	865.72	0	2501	948.00	1353.00	1535.00	2322.12	0.61	2.67	0.41
Purchase 01	258	29	11.24%	875.49	0	2350	976.50	1335.60	1437.55	2313.41	0.63	3.82	0.49
Purchase 02	242	26	10.74%	876.26	0	2409	939.50	1334.30	1514.00	2337.13	0.59	3.60	0.46
Purchase 03	262	26	9.92%	883.65	0	2502	944.50	1297.10	1500.70	2493.59	0.64	3.92	0.50
Reservation 01	252	23	9.13%	893.16	0	2501	945.50	1338.10	1497.25	2484.22	0.61	4.13	0.46
Reservation 02	262	28	10.69%	899.36	0	2425	979.50	1304.10	1480.65	1957.49	0.64	4.24	0.47
Reservation 03	243	19	7.82%	894.12	0	2496	934.00	1342.60	1505.20	2478.00	0.59	4.03	0.46

Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Confirmation 01	439.734211	887.037037	2489	0
Confirmation 02	466.5775647	948.7306122	2495	0
Confirmation 03	435.7712535	900.2661597	2500	0
Home page 01	419.7951934	859.6272401	2351	0
Home page 02	424.5899778	876.6862745	2478	0
Home page 03	450.9204062	855.749004	2501	0
Purchase 01	429.4544096	853.7945736	2326	0
Purchase 02	428.7969008	857.9545455	2409	0
Purchase 03	425.4475129	859	2496	0
Reservation 01	428.479122	869.7103175	2501	0
Reservation 02	418.3072363	880.7061069	2424	0
Reservation 03	424.2455382	870.9876543	2495	0
Grand Total	432.5878506	876.4212766	2501	0

Observations:

1. During stress test error occurred which is **9.98%** (Non-HTTP response code).
2. Average Response time **893.26ms**
3. 95th percentile Response times - **1486.40ms**
4. Throughput (requests per second) - **7.44ms**
5. Network performance (Max latency) – **2501ms**
6. Network performance (Average latency) – **948.73ms**

Spike Test

In this spike test, which is conducted by open model thread group to perform 1000 users interact within 10 seconds. In this test started by 10 seconds and completed by 40 seconds. Load rate will be 400/seconds. Around 8900 samples were transitioned.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	8900	73	0.82%	3733.77	43	50423	3005.00	8208.90	9196.75	10557.90	132.57	790.29	114.51
Confirmation 01	698	10	1.43%	3863.31	54	10731	3199.00	8334.20	9534.65	10637.18	10.40	57.99	10.28
Confirmation 02	709	9	1.27%	3781.26	77	13158	3007.00	8183.00	9373.50	10670.80	10.56	58.95	10.39
Confirmation 03	761	6	0.79%	3725.54	76	10759	3159.00	8280.00	8922.80	10524.28	11.34	63.61	11.16
Home page 01	761	5	0.66%	3810.62	89	10841	3130.00	8281.20	9344.90	10626.24	11.34	52.44	8.73
Home page 02	774	5	0.65%	3678.01	43	10679	3139.50	8000.50	8945.25	10382.25	11.53	53.34	8.88
Home page 03	743	5	0.67%	3654.18	90	10759	2861.00	8412.60	9297.00	10400.36	11.07	51.14	8.54
Purchase 01	775	8	1.03%	3794.24	44	10945	3136.00	8129.80	9125.00	10502.04	11.55	75.10	10.12
Purchase 02	735	5	0.68%	3720.00	51	11089	3010.00	8189.00	9023.00	10240.04	10.95	71.43	9.53
Purchase 03	720	5	0.69%	3752.95	106	50423	2895.00	8326.00	9492.55	10555.08	10.72	69.97	9.37
Reservation 01	758	6	0.79%	3681.47	44	10817	2884.00	8303.20	9477.30	10649.66	11.30	80.46	9.37
Reservation 02	743	5	0.67%	3736.02	58	10798	2960.00	8442.60	9458.20	10591.76	11.07	78.85	9.17
Reservation 03	723	4	0.55%	3614.00	69	10715	2742.00	7952.60	9120.60	10545.56	10.77	77.17	8.97

Type of error	Number of errors	% In errors	% In all samples
429/Too Many Requests	50	68.49%	0.56%
Non HTTP response code: java.net.SocketException/Non HTTP response message: Socket closed	23	31.51%	0.26%

Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Confirmation 01	3018.731857	3848.758	10719	0
Confirmation 02	2959.974399	3766.299	13144	0
Confirmation 03	2895.893135	3706.472	10755	0
Home page 01	2968.007764	3794.57	10834	0
Home page 02	2843.969618	3663.486	10669	0
Home page 03	2953.399397	3641.082	10756	0
Purchase 01	2963.12984	3764.686	10920	0
Purchase 02	2902.25463	3689.566	11063	0
Purchase 03	2974.306485	3656.814	10954	0
Reservation 01	2964.214773	3654.723	10811	0
Reservation 02	2972.086916	3706.205	10772	0
Reservation 03	2918.648756	3582.138	10700	0
Grand Total	2943.210703	3705.858	13144	0

Observations:

1. During Spike test error occurred which is **0.82%** (Non-HTTP response code & 429 error)
2. Average Response time **3733.77ms**
3. 95th percentile Response times - **9196.75ms**
4. Throughput (requests per second) - **132.57ms**
5. Network performance (Max latency) – **13144ms**
6. Network performance (Average latency) – **3848.75ms**

Endurance Test

In this endurance test, which is conducted by Concurrency thread group to perform 200 users interact within 90 minutes. In this test, the full load started after 60 seconds and was completed by 1 hour 41 minutes. Duration assertion included for 3 minutes. Each request is integrated with a uniform random timer to simulate real-world scenarios. All the requests will under synchronize timer for 200 concurrent users.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	187102	519	0.28%	498.18	8	47525	480.00	905.00	1052.95	1380.00	30.88	185.02	26.71
Confirmation 01	15701	43	0.27%	496.40	11	10397	450.00	587.00	729.00	1133.98	2.59	14.61	2.56
Confirmation 02	15658	41	0.26%	493.12	17	9094	449.00	573.00	712.05	1137.82	2.58	14.57	2.54
Confirmation 03	15679	44	0.28%	495.26	14	10263	449.00	583.00	731.00	1167.20	2.59	14.58	2.55
Home page 01	15577	44	0.28%	493.35	26	10578	443.00	575.00	715.00	1143.44	2.57	11.93	1.98
Home page 02	15762	43	0.27%	490.06	20	10496	443.00	574.00	708.00	1142.11	2.60	12.07	2.00
Home page 03	15603	43	0.28%	491.48	14	21500	443.00	572.00	710.00	1154.92	2.58	11.95	1.98
Purchase 01	15451	47	0.30%	502.10	19	9875	461.00	592.00	729.00	1146.00	2.55	16.69	2.24
Purchase 02	15659	45	0.29%	505.73	15	10470	462.00	592.00	731.00	1137.40	2.58	16.92	2.25
Purchase 03	15502	47	0.30%	508.07	8	15541	462.00	600.00	744.85	1191.91	2.56	16.75	2.23
Reservation 01	15415	41	0.27%	501.15	21	47525	457.00	582.40	724.00	1128.84	2.54	18.21	2.11
Reservation 02	15559	34	0.22%	503.30	17	21832	458.00	589.00	735.00	1143.00	2.57	18.36	2.13
Reservation 03	15536	47	0.30%	498.36	15	21923	458.00	597.00	739.15	1112.00	2.56	18.41	2.13

Type of error	Number of errors	% in errors	% in all samples
Non HTTP response code: java.net.BindException/Non HTTP response message: Address already in use: connect	512	98.65%	0.27%
429/Too Many Requests	4	0.77%	0.00%
Non HTTP response code: java.net.SocketException/Non HTTP response message: Connection reset	3	0.58%	0.00%

Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Confirmation 01	276.7488553	486.5291	10397	0
Confirmation 02	251.7154696	483.5582	9090	0
Confirmation 03	260.279774	485.649	10250	0
Home page 01	335.356299	483.6319	10568	0
Home page 02	289.2023367	480.4071	10496	0
Home page 03	346.2142639	481.5846	21490	0
Purchase 01	232.5247716	480.7476	9849	0
Purchase 02	279.1602181	484.4655	10470	0
Purchase 03	309.617602	487.0657	15534	0
Reservation 01	453.7591761	479.4559	47521	0
Reservation 02	374.7292532	481.6689	21826	0
Reservation 03	251.8775553	476.0026	21913	0
Grand Total	310.8915934	482.5708	47521	0

Observations:

1. During Endurance test error occurred which is **0.28%** (Non-HTTP response code & 429 error)
2. Average Response time **498.18ms**
3. 95th percentile Response times - **1052.95ms**
4. Throughput (requests per second) – **30.88ms**
5. Network performance (Max latency) – **47521ms**
6. Network performance (Average latency) – **486.52ms**

Volume Test

In this volume test, which is conducted by the Ultimate thread group to perform 800 users interact with 300 seconds. In this test, which was performed by various load actives 100, 300, 500, and full load 800. This full load 800 users in 15 minutes. Duration assertion included for 3 minutes to assertion of which over 3 minutes of transactions. Each request is integrated with a uniform random timer and random controller to simulate real-world scenarios.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	107777	14378	13.34%	605.85	6	22960	586.00	955.00	1103.00	1700.99	88.86	491.24	66.92
Confirmation 01	8983	1213	13.50%	598.14	12	4895	500.00	924.00	1120.00	1971.96	7.46	38.91	6.41
Confirmation 02	9149	1237	13.52%	596.49	7	22223	497.00	923.00	1123.00	1996.00	7.60	39.62	6.49
Confirmation 03	8908	1198	13.45%	611.92	11	22414	502.00	931.10	1144.55	1998.64	7.35	38.34	6.30
Home page 01	9032	1225	13.56%	601.40	12	21709	496.00	920.00	1122.00	2166.03	7.51	32.62	5.02
Home page 02	8954	1212	13.54%	598.81	13	21987	494.00	908.00	1109.25	1943.35	7.42	32.26	4.97
Home page 03	9056	1183	13.06%	597.30	14	22195	494.00	912.00	1093.00	1879.43	7.51	32.74	5.06
Purchase 01	9078	1233	13.58%	610.22	12	22960	509.00	935.00	1114.00	1905.52	7.53	45.20	5.74
Purchase 02	8981	1138	12.67%	615.33	11	22149	511.00	939.80	1133.90	2035.34	7.45	45.00	5.69
Purchase 03	8880	1211	13.64%	612.89	11	4881	517.00	951.00	1136.95	1940.76	7.36	44.13	5.58
Reservation 01	8933	1214	13.59%	615.18	11	22292	508.00	923.00	1146.30	2229.64	7.41	48.35	5.33
Reservation 02	9005	1183	13.14%	604.75	6	21663	506.00	927.00	1115.70	1991.00	7.46	48.83	5.40
Reservation 03	8818	1131	12.83%	608.17	13	22660	510.00	917.00	1126.00	1927.11	7.32	48.16	5.33

Row Labels	StdDev of Latency	Average of Latency	Max of Latency	Min of Latency
Confirmation 01	357.9532162	528.2732	4890	0
Confirmation 02	419.8196383	525.9235	22223	0
Confirmation 03	574.8287715	537.8172	22414	0
Home page 01	478.7394	528.8553	21694	0
Home page 02	524.5517491	528.0707	21977	0
Home page 03	470.0884406	525.8652	22188	0
Purchase 01	475.3394014	527.3417	22959	0
Purchase 02	425.07716	536.8269	22122	0
Purchase 03	347.0643985	527.0019	4857	0
Reservation 01	533.1298211	530.4066	22292	0
Reservation 02	414.8837071	522.459	21637	0
Reservation 03	475.6231414	529.3324	22660	0
Grand Total	462.6473826	528.9995	22959	0

Observations:

1. During Volume test error occurred which is **13.34%**.
2. Average Response time **605.85ms**
3. 95th percentile Response times - **1103ms**
4. Throughput (requests per second) – **88.86ms**
5. Network performance (Max latency) – **22959ms**
6. Network performance (Average latency) – **537.81ms**
7. Error starts after **64757 transactions** out of 107778 transactions.

Conclusion

After conducting various tests from various thread groups, we will come to know

- **Load Test** - After performing load test we received **7.67%** of error around 1.34 lakh transactions. Error starts after 12500 requests processed.
- **Stress Test** - After performing stress test we received **9.98%** of error around 3055 transactions. Error starts after 2752 requests processed Latency Report shows confirmation page makes higher latency. (Non-HTTP response message: Network is unreachable: connect).
- **Endurance Test** - After performing Endurance test, we received **0.28%** of error. Major error occurs at non http respond. Reservation page receives higher latency.
- **Spike Test** - After performing Spike test, we received **0.82%** of error. Average latency in maximum receives at confirmation page.
- **Volume Test** - After performing the volume test, an error occurred, which is **13.34%**. Maximum error occurs due to duration assert, which is caused by responses taking more than 3 seconds.

Website can able to handle Endurance and Spike Tests. But its unable to handle consistently while load increased gradually. Stability of load test is very better than other pages.

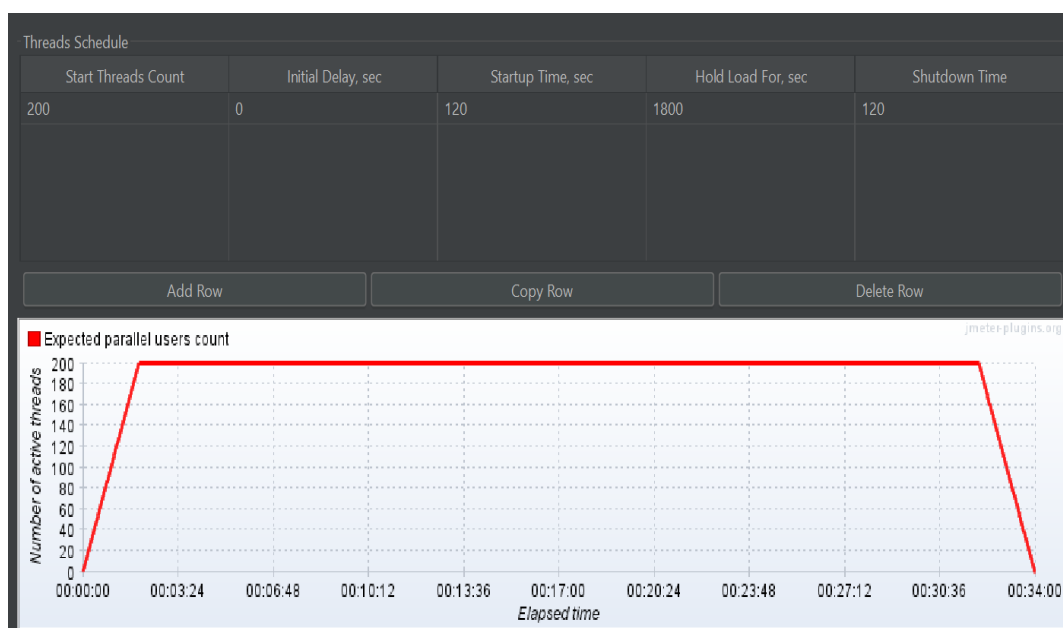
Appendix

Load Test



Configurations

- **Assertion** - Duration Assertion 2 seconds for 90% & 3 seconds 95%.
- **Timer** - Synchronizing Timer
- **Thread Group** - Ultimate Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



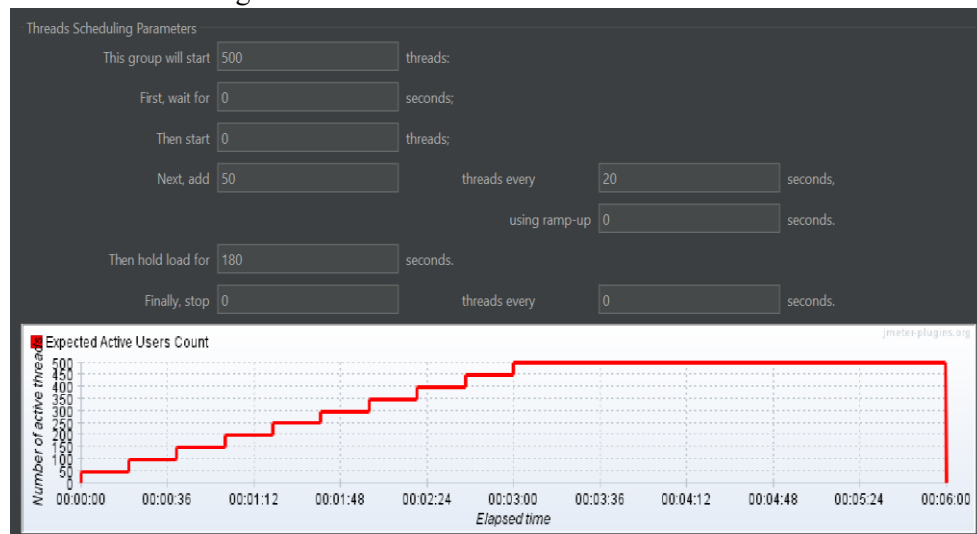
Stress Test



Blazemeter Stress
test.jmx

Configurations

- **Assertion** - Duration Assertion 3 seconds 95%.
- **Timer** - Synchronizing Timer
- **Thread Group** - Stepping Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



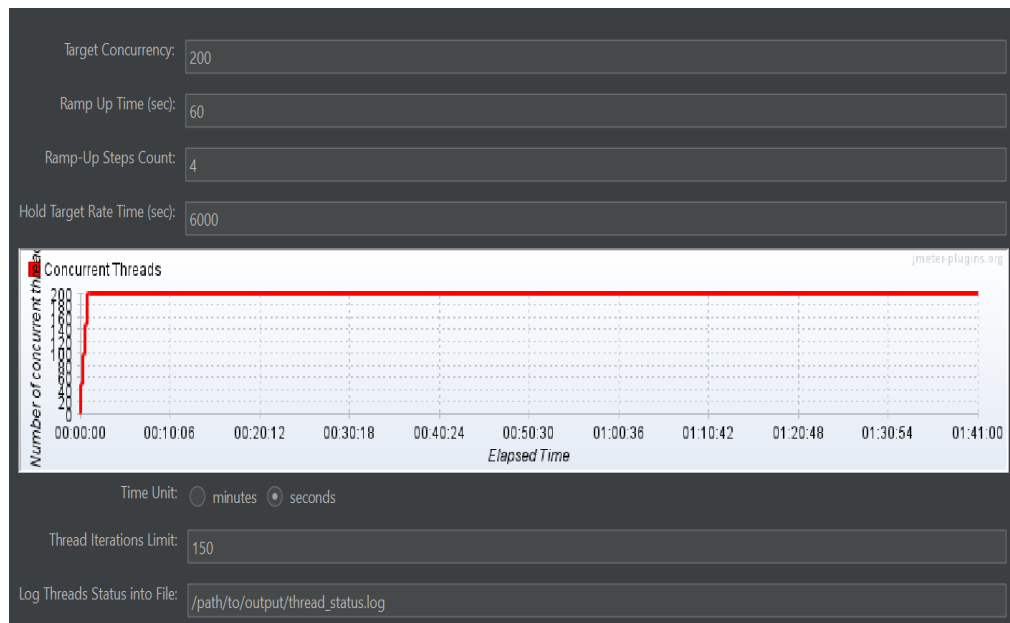
Endurance Test



Blazemeter
Endurance test.jmx

Configurations

- **Assertion** - Duration Assertion 3 seconds 95%.
- **Timer** - Synchronizing Timer, Uniform Random Timer
- **Thread Group** - Concurrency Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



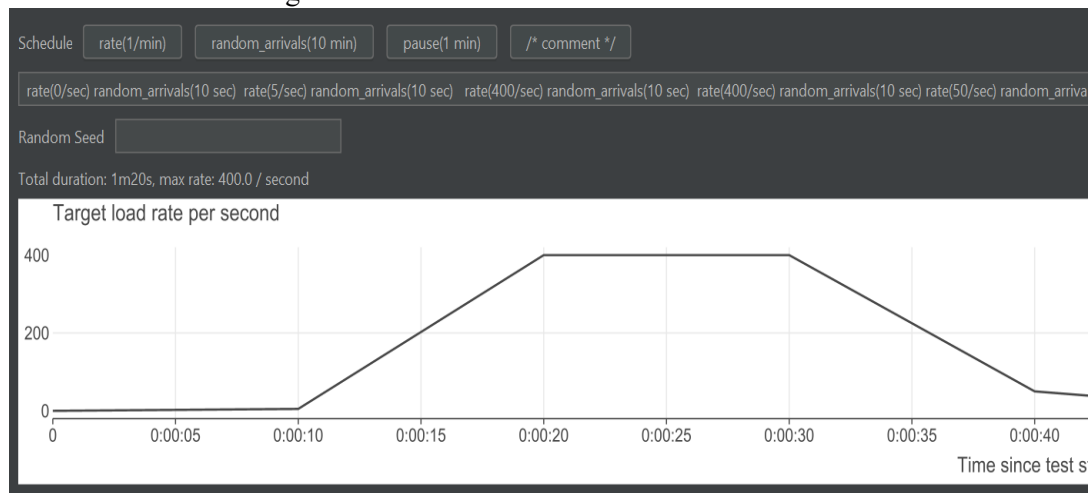
Spike Test



Blazedemo Spike
test.jmx

Configurations

- **Assertion** – No Assertion Integrated
- **Timer** - Synchronizing Timer
- **Thread Group** – Open Model Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



Volume Test



Blazemeter Scalability
test.jmx

Configurations

- **Assertion** - Duration Assertion 3 seconds.
- **Timer** - Uniform Random Timer
- **Thread Group** - Ultimate Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager

